INCH-POUND
J-W-1177/15B
June 10, 1988
SUPERSEDING
J-W-1177/15A
September 27, 1976

FEDERAL SPECIFICATION SHEET

WIRE, MAGNET, ELECTRICAL, CLASS 220, TYPE M, POLYIMIDE-COATED, ROUND

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

The requirements for acquiring the wire described herein shall consist of this specification and the latest issue of J-W-1177.

Classification: Class 220; type M (single), type M2 (heavy), type M3 (triple), type M4 (quadruple); round.
Insulating materials: The film shall be based on polyimide resin.
NEMA/ANSI equivalent: M, M2, and M3 test requirements, except for thermal endurance and certain wire sizes required for stock standardization, are equivalent to MW-16 of NEMA MW 1000. M4 is not specified in NEMA MW 1000.
General requirements: See J-W-1177 for general requirements, quality assurance provisions, and packaging.
Requirements:

Characteristics	Test procedure, see J-W-1177	Wire sizes, AWG	Requirements
Dimensions	4.7.1.2	4-50	See tables I and II.
Adherence and flexibility	4.7.2.1	4-50	No cracks visible in the coating.
Elongation	4.7.5	4-50	Not less than the value in table III.
Heat shock	4.7.4	4-44	No cracks visible in the coating after conditioning as shown in table IV.

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Requirements: (Continued)

Characteristics	Test procedure, see J-W-1177	Wire sizes, AWG	Requirements
Scrape resistance	4.7.6	10-30	Lowest grams-to-fail load for any of the three tests and the average of the three tests shall be not less than the values in table V.
Springback	4.7.7	14-30	Not greater than the value in table VI.
Dielectric strength	4.7.9	4-44	Not less than the value in table VII.
Continuity	4.7.10	31-50	The number of discontinuities shall be not greater than the number listed in table VIII.
	4.7.11	14-30	
Completeness of cure	4.7.16.2	4-50	Dissipation factor shall be not greater than 0.60 percent.
Thermoplastic flow	4.7.8	18, 36	Median shall be not less than 400°C with heavy film coated wire.
Solubility	4.7.12	18, 36	Heavy film coated wire shall not soften sufficiently to expose bare conductor when immersed in xylene or 50/50 parts by volume xylene/ethyl Cellosolve.
Dielectric strength at temperature	4.7.14	18, 36	Heavy film coated wire shall average not less than 4275 volts for 18 AWG or 1900 volts for 36 AWG.
Thermal endurance	4.7.15.1	18	220°C minimum with heavy film coated wire.
	4.7.15.2	4-44	2000 volts/mil minimum after 168 hours at 275°C.
	4.7.15.3	4-44	240°C minimum as shown in table IV.

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TABLE I. Dimensions, sizes 4 to 44 AWG.

AWG size	Bare wire diameter, inch			Type M, single		Type M2, heavy		Type M3, triple		Type M4, quadruple	
	Minimum	Nominal	Maximum	Minimum increase in diameter, inch	Maximum overall diameter, inch	Minimum increase in diameter, inch	Maximum overall diameter, inch	Minimum increase in diameter, inch	Maximum overall diameter, inch	Minimum increase in diameter, inch	Maximum overall diameter, inch
1/ 4	0.2023	0.2043	0.2053	0.0018	0.2079	0.0037	0.2098	0.0056	0.2117	0.0074	0.2135
1/ 5	.1801	.1819	.1828	.0018	.1854	.0036	.1872	.0054	.1890	.0072	.1908
1/ 6	.1604	.1620	.1628	.0017	.1653	.0035	.1671	.0052	.1688	.0070	.1706
1/ 7	.1429	.1443	.1450	.0017	.1474	.0034	.1491	.0051	.1508	.0068	.1525
1/ 8	.1272	.1285	.1292	.0016	.1315	.0033	.1332	.0049	.1348	.0066	.1365
1/ 9	.1133	.1144	.1150	.0016	.1173	.0032	.1189	.0048	.1205	.0064	.1221
1/10	.1009	.1019	.1024	.0015	.1045	.0031	.1061	.0046	.1076	.0062	.1092
1/11	.0898	.0907	.0912	.0015	.0933	.0030	.0948	.0045	.0963	.0060	.0978
1/12	.0800	.0808	.0812	.0014	.0832	.0029	.0847	.0043	.0861	.0058	.0876
1/13	.0713	.0720	.0724	.0014	.0743	.0028	.0757	.0042	.0771	.0056	.0785
1/14	.0635	.0641	.0644	.0016	.0666	.0032	.0682	.0048	.0700	.0064	.0715
1/15	.0565	.0571	.0574	.0015	.0594	.0030	.0609	.0045	.0627	.0060	.0644
1/16	.0503	.0508	.0511	.0014	.0531	.0029	.0545	.0043	.0562	.0057	.0577
1/17	.0448	.0453	.0455	.0014	.0475	.0028	.0488	.0041	.0504	.0055	.0520
1/18	.0399	.0403	.Q405	.0013	.0424	.0026	.0437	.0039	.0452	.0052	.0468
1/19	.0355	.0359	.0361	.0012	.0379	.0025	.0391	.0037	.0406	.0049	.0422
1/20	.0317	.0320	.0322	.0012	.0339	.0023	.0351	.0035	.0364	.0047	.0379
1/21	.0282	.0285	.0286	.0011	.0303	.0022	.0314	.0033	.0326	.0044	.0342
1/22	.0250	.0253	.0254	.0011	.0270	.0021	.0281	.0032	.0293	.0043	.0308
1/23	.0224	.0226	.0227	.0010	.0243	.0020	.0253	.0030	.0264	.0040	.0279
1/24	.0199	.0201	.0202	.0010	.0217	.0019	.0227	.0029	.0238	.0039	.0252
1/25	.0177	.0179	.0180	.0009	.0194	.0018	.0203	.0027	.0214	.0036	.0228
1/26	.0157	.0159	.0160	.0009	.0173	.0017	.0182	.0026	.0193	.0035	.0206
1/27	.0141	.0142	.0143	.0008	.0156	.0016	.0164	.0024	.0173	.0032	.0185
1/28	.0125	.0126	.0127	.0008	.0140	.0016	.0147	.0023	.0156	.0031	.0166
1/29	.0112	.0113	.0114	.0007	.0126	.0015	.0133	.0022	.0142	.0029	.0152
1/30	.0099	.0100	.0101	.0007	.0112	.0014	.0119	.0021	.0128	.0028	.0137

See footnote at end of table.

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TABLE I. Dimensions, sizes 4 to 44 AWG. - Continued

AWG size	Bare wire diameter, inch			Type M, single		Type M2, heavy		Type M3, triple		Type M4, quadruple	
	Minimum	Nominal	Maximum	Minimum increase in diameter, inch	Maximum overall diameter, inch	Minimum increase in diameter, inch	Maximum overall diameter, inch	Minimum increase in diameter, inch	Maximum overall diameter, inch	Minimum increase in diameter, inch	Maximum overall diameter, inch
31	0.0088	0.0089	0.0090	0.0006	0.0100	0.0013	0.0108	0.0019	0.0114	0.0026	0.0121
32	.0079	.0080	.0081	.0006	.0091	.0012	.0098	.0018	.0104	.0024	.0110
33	.0070	.0071	.0072	.0005	.0081	.0011	.0088	.0016	.0093	.0022	.0099
34	.0062	.0063	.0064	.0005	.0072	.0010	.0078	.0015	.0083	.0020	.0088
35	.0055	.0056	.0057	.0004	.0064	.0009	.0070	.0013	.0074	.0018	.0079
36	.0049	.0050	.0051	.0004	.0058	.0008	.0063	.0012	.0067	.0016	.0071
37	.0044	.0045	.0046	.0003	.0052	.0008	.0057	.0012	.0061	.0016	.0065
38	.0039	.0040	.0041	.0003	.0047	.0007	.0051	.0010	.0054	.0014	.0058
39	.0034	.0035	.0036	.0002	.0041	.0006	.0045	.0009	.0048	.0012	.0051
40	.0030	.0031	.0032	.0002	.0037	.0006	.0040	.0009	.0043	.0012	.0046
41	.0027	.0028	.0029	.0002	.0033	.0005	.0036	.0007	.0038	.0010	.0041
42	.0024	.0025	.0026	.0002	.0030	.0004	.0032	.0006	.0034	.0008	.0036
43	.0021	.0022	.0023	.0002	.0026	.0004	.0029	.0006	.0031	.0008	.0033
44	.0019	.0020	.0021	.0001	.0024	.0004	.0027	.0006	.0029	.0008	.0031

1/ These bare wire diameters may be exceeded, provided:

(a) The maximum diameters specified by QQ-W-343 are not exceeded,
(b) The minimum increases in diameter shown in table I are maintained, and
(c) The maximum overall diameters shown in table I are not exceeded.

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TABLE II. Characteristics of wire, sizes 45 to 50 AWG.

AWG size	Theoretical 1/ nominal bare wire diameter, inch	Conductor resistance at 20°C ohms per foot			Type M, single		Type M2, heavy	
		Minimum	Nominal	Maximum	Minimum increase in diameter, inch	Maximum overall diameter, inch	Minimum increase in diameter, inch	Maximum overall diameter, inch
45	0.00176	3.080	3.348	3.616	0.00010	0.00205	0.00030	0.00230
46	.00157	3.870	4.207	4.544	.00010	.00185	.00030	.00210
47	.00140	4.868	5.291	5.714	.00010	.00170	.00030	.00190
48	.00124	6.205	6.745	7.285	.00010	.00150	.00020	.00170
49	.00111	7.744	8.417	9.090	.00010	.00130	.00020	.00150
50	.00099	9.734	10.58	11.43	.00010	.00120	.00020	.00140

1/ Theoretical nominal bare wire diameters are in accordance with NBS Handbook 100. Conductor diameter tolerances are shown as resistance values and shall be determined by measuring the resistance of the wire in accordance with ASTM B 193, where applicable. A specimen at least 5 feet long shall be used.

TABLE III. Elongation of finished wire.

AWG size	Minimum elongation, percent	AWG size	Minimum elongation, percent	AWG size	Minimum elongation, percent	AWG size	Minimum elongation, percent	AWG size	Minimum elongation, percent
4	38	14	33	24	28	33	23	42	16
5	37	15	33	25	28	34	22	43	15
6	37	16	33	26	27	35	21	44	14
7	36	17	32	27	27	36	20	45	11
8	36	18	32	28	26	37	20	46	10
9	36	19	31	29	26	38	19	47	8
10	35	20	30	30	25	39	18	48	7
11	35	21	30	31	24	40	17	49	6
12	34	22	29	32	24	41	17	50	5
13	34	23	29						

TABLE IV. Heat shock.

AWG size	Minimum elongation, percent	Mandrel diameter	Minimum temperature, °C
4-9	30	None	240
10-13	25	5X	240
14-30	20	3X	240
31-44	1/20	3X	240

1/ Or to the breaking point, whichever is less.

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TABLE V. Scrape resistance.

AWG size	Type M		Type M2		Type M3		Type M4	
	Average grams-to-fail	Minimum grams-to-fail	Average grams-to-fail	Minimum grams-to-fail	Average grams-to-fail	Minimum grams-to-fail	Average grams-to-fail	Minimum grams-to-fail
10	545	465	1090	925	1360	1150	1635	1385
11	545	465	1090	925	1360	1150	1635	1385
12	545	465	1090	925	1360	1150	1635	1385
13	545	465	1090	925	1360	1150	1635	1385
14	545	465	1090	925	1360	1150	1635	1385
15	510	430	1015	865	1270	1080	1520	1295
16	475	405	955	810	1190	1010	1430	1215
17	445	380	890	755	1115	945	1335	1130
18	420	355	835	710	1045	890	1250	1065
19	390	335	785	665	980	830	1175	995
20	370	315	740	630	920	785	1110	945
21	345	295	695	595	865	735	1040	890
22	325	275	650	550	810	690	975	825
23	300	255	605	515	755	645	905	770
24	285	245	565	485	710	605	845	725
25	265	225	530	450	665	565	795	675
26	245	210	495	425	620	530	740	635
27	230	200	460	395	575	490	860	590
28	215	185	435	370	540	460	650	555
29	200	170	405	340	505	425	605	510
30	190	160	380	325	470	395	570	485

TABLE VI. Springback of finished wire.

AWG size	Maximum springback, degrees		AWG size	Maximum springback, degrees	
	Types M and M2	Types M3 and M4		Types M and M2	Types M3 and M4
14	42	46	23	62	64
15	46	49	24	67	69
16	50	53	25	72	74
17	54	58	26	76	80
18	58	62	27	50	55
19	62	66	28	55	60
20	66	70	29	61	65
21	53	53	30	66	70
22	58	58			

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TABLE VII. Minimum breakdown voltages.1/

AWG size	Volts			AWG size	Volts		
	Type M	Type M2	Type M3		Type M	Type M2	Type M3
4	2055	3700	4930	25	2625	4725	6325
5	2000	3600	4800	26	2550	4600	6150
6	1945	3500	4670	27	2500	4500	6000
7	1885	3400	4525	28	2425	4375	5850
8	1835	3300	4405	29	2375	4250	5700
9	1775	3200	4260	30	2300	4150	5550
10	3445	6200	8268	31	2075	3825	4850
11	3335	6000	8005	32	1850	3525	4725
12	3220	5800	7730	33	1675	3250	4500
13	3110	5600	7465	34	1500	2975	4250
14	3525	6325	8450	35	1325	2750	3825
15	3425	6175	8225	36	1200	2525	3525
16	3325	6000	8000	37	1075	2325	3525
17	3250	5850	7800	38	950	2150	2975
18	3175	5700	7600	39	850	1975	2750
19	3075	5550	7400	40	775	1800	2750
20	3000	5400	7200	41	700	1675	2150
21	2925	5250	7025	42	625	1525	1975
22	2850	5125	6850	43	550	1400	1975
23	2775	5000	6675	44	500	1300	1975
24	2700	4850	6500				

1/ All sizes of type M4 shall be not less than 2000 volts/mil.

TABLE VIII. Continuity.

AWG size	Maximum number of discontinuities			
	Type M	Type M2	Type M3	Type M4
14-24	25	5	3	1
25-30	25	7	5	3
31-46	25	5	3	1
47-50	25	10	7	5

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Part number: Magnet wire covered by this specification shall be defined by the following part numbering system. Example: M1177/15-02C029

M1177/15-	02	C	029
Federal specification identifier	Two digit type code	Single letter conductor code	Three character size code

The following codes shall apply:

Type	Type code	Conductor	Conductor code
M	01	Copper	C
M2	02	Aluminum	A
M3	03	Nickel-coated copper	N
M4	04	Silver-coated copper	S

The size code shall be the bare wire dimension. AWG wire size shall be used.

Intended use: Type M magnet wire is intended for use in 220°C applications similar to those for which type T is used. Type M magnet wire has been standardized for the repair of shipboard power equipment.

Revision letters are not used to denote changes due to the extensiveness of the changes.

MILITARY INTERESTS:

Custodians:
Army - CR
Navy - SH
Air Force - 85

Review activities:
Army - AR, ER, MI
DLA - IS

User activities:
Army - ME
Navy - AS, CG, MC, OS

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS, PBO, PCD
INTERIOR - BLM
HHS - FDA
DCGOVT - DCG
NASA - JFK
COMMERCE - NBS
TRANSPORTATION - APM, FAA

Preparing activity:
Navy - SH
(Project 6145-1111-12)

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